PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	The effectiveness of meatal cleaning in the prevention of catheter-associated urinary tract infections and bacteriuria: An updated systematic review and meta-analysis
AUTHORS	Mitchell, Brett; Curryer, Cassie; Holliday, Elizabeth; Rickard, Claire; Fasugba, Oyebola

VERSION 1 – REVIEW

REVIEWER	Solis-Trapala, I Keele, Institute for Applied Clinical Sciences
REVIEW RETURNED	31-Jan-2021

GENERAL COMMENTS	Review of "The effectiveness of meatal cleaning in the prevention of catheter-associated urinary tract 3 infections and bacteriuria: An updated systematic review and meta-analysis" The manuscript is written clearly and concisely. Below are some comments and suggestions. Abstract and Introduction: Please include information on the study participants and study eligibility criteria The two first statements under heading "Strengths and limitations of this study" sound equivalent to me. Methods: P 3 L91 Please specify the approach or aspects to the approach referred to Please provide some rationale on the selection of databases that ensures the comprehensiveness of the systematic review. The authors undertook single instead of double data extraction for the systematic review. This is a study limitation that should be acknowledged Statistical analysis The sizes of the strata in the stratified analyses, 1-4 (Fig. 2, S1, S2, S3), are too small to conduct random-effects meta-analysis as inference in random-effects models requires a larger number of studies included in meta-analysis to guarantee reliable conclusions. Results and Discussion: Table S1, P 15 L 260 and 275: P-values alone do not provide an informative description of effects (Table S1), effect estimates and confidence intervals should be included.

REVIEWER	Laan, Bart Amsterdam UMC - Locatie AMC, Internal Medicine, Infectious Diseases
REVIEW RETURNED	02-Feb-2021

GENERAL COMMENTS	Thank you for the opportunity to read this paper describing the results of a systematic review. A well written interesting paper.

	-I am not completely convinced of the effect of meatal cleaning prior to urinary catheterisation to reduce CAUTI. This outcome is addressed only in two studies and one of those showed no differences. Could the authors address this and other limitations in the discussion section? -Basically, almost all results are non significant. Still, the authors stated that the results are positive and have clinical impact. This should be addressed in more details. What is this clinical impact? Since reducing just bacteriuria have no clinical impact. -The outcome could be addressed more clearly.

VERSION 1 – AUTHOR RESPONSE

	Feedback	**Comments**	**Action**
	Reviewer 1		
1	Please include information on the study participants and study eligibility criteria (abstract)	Agree	Included in abstract (introduction)
2	The two first statements under heading “Strengths and limitations of this study” sound equivalent to me	Agree	Removed second dot point. Included a comment regarding sub-group analysis
3	P 3 L91 Please specify the approach or aspects to the approach referred to	Agree. Clarification added.	Included “the methodological approach used in this systematic review is the same as…”
4	Please provide some rationale on the selection of databases that ensures the comprehensiveness of the systematic review.	These are databases commonly used in systematic reviews. They are consistent with the previous systematic review, such that the updated reviewed is consistent.	No action

5	The authors undertook single instead of double data extraction for the systematic review. This is a study limitation that should be acknowledged	Agree. Added as a limitation	Limitations added into the discussion section.
6	The sizes of the strata in the stratified analyses, 1-4 (Fig. 2, S1, S2, S3), are too small to conduct random-effects meta-analysis as inference in random-effects models requires a larger number of studies included in meta-analysis to guarantee reliable conclusions.	We have carefully considered the reviewers feedback on this, reviewed the literature and sought additional advice from methodologists at Cochrane Australia. This is in addition to the expertise of the authors which includes a biostatistician. Based on these, in this instance, we mostly agree with the reviewer's comments and have made some changes to incorporate key elements. Our rationale includes: -The population groups in stratified analysis differ, thus potentially violating assumptions for using a fixed effect model. This needs to	We have added as a limitation the heterogeneity of studies in Figure 2 in the limitation section. We have used a fixed effect model for Figures S1 to S3 inclusive. Included this the methods.

		weighed against sub-analysis that has limited studies. -Random effects should be used in the first instance, with a strong justification for a fixed effect mode. -The intent of this systematic review, as articulated in the abstract, methods and introduction is to update a previous review, such that new studies can be included. For these reasons, we feel it is important to replicate the methods used in the original study – at least for Figure 2. -Figure 2 is produced in manner consistent with the original publication which we feel is critical, given this is an update and used for comparison. -A random-effects model is likely to provide wider confident intervals and thus provide a greater appreciation of the potential uncertainly for results in stratified analysis Noting the above and points made by the	We have tempered some discussion and included headings with the aim of improving clarity. We have provided an additional sentence in the conclusion.

		reviewer, we have -Left Figure 2 as using a random-effects, noting the larger number of studies in this meta-analysis -Modified and presented data for stratified analysis to use a fixed effect method, with random-effects also presented in supplementary data for transparency. It is worth noting that by using a fixed effect, one outcome (chlorhexidine on CAUTI) is now statistically significant. We still feel the need to temper this result somewhat, noting the above and the comments from reviewer 2 (which were based on a random-effects model).	
7	Table S1, P 15 L 260 and 275: P-values alone do not provide an informative description of effects (Table S1), effect estimates and confidence intervals should be included.	Agree and we have included the CIs. P values have been removed in the section referred to in the discussion. With regards to Table S1, the p values presented supplement the data presented in Figure S1, where odds ratios and confidence intervals are provided (but no p value as this is a system generated figure). The intent was not to present these are stand alone, but rather to supplement	Removal of p values in discussion. Addition of text in supplementary information.
		Figure S1. We have made a note under Table S1	
		to improve clarity and intent.	

	Reviewer 2		
8	I am not completely convinced of the effect of meatal cleaning prior	We agree and have addressed this in a number of ways: -Additional information in the discussion to	Additions to discussion

	to urinary catheterisation to reduce CAUTI…Could the authors address this and other limitations in the discussion section?	temper the language and provide caveats -Provide some additional information on the direction of studies -Included headings in the results and discussion for improved clarity and differentiate key results by outcome (bacteriuria and CAUTI) -Added issues related to heterogeneity in a new 'limitations' section. -Added a sentence in the "The use of antiseptics prior to urinary catheterisation and effect on CAUTI" section in the discussion, to pick up an important point re differences in studies.	Addition of headings in discussion and results. Additional sentences added in the discussion at the end of the section "use of antiseptics prior to urinary catheterisation and effect on CAUTI. Addition of limitation heading and key points. Additional sentence in conclusion.
9	Basically, almost all results are non significant. Still, the authors stated that the results are positive and	Agree. Please see above. With regards to bacteriuria, we don't disagree re clinical impact. However, bacteriuria is commonly (mis) treated as infection. In the era of	Please see above. Additionally, section added in

	have clinical impact. This should be addressed in more details. What is this clinical impact? Since reducing just bacteriuria have no clinical impact.	needing to reducing overall antimicrobial use, reducing bacteriuria will have a positive effect in the context of reducing inappropriate prescribing. We have also added commentary in the discussion indicating that bacteriuria is not an appropriate outcome – but one that is largely used. Noting the points above, we have also included headings, such that there is a clear section discussing findings related to CAUTI (infection). We have included headings in the results and discussion for improved clarity and differentiate key results by outcome (bacteriuria and CAUTI)	the discussion at the end of the section “The effect of meatal cleaning in reducing the risk of bacteriuria”. Additional sentences in the conclusion.

		Using a fixed effect model, the results for using chlorhexidine on infection as the outcome (CAUTI) are now statistically significant. The use of a fixed effect model was at the request of reviewer 1. Therefore, results while needing to be tempered, are statistically significant for this subgroup. We has three sentences in the section “Use of antiseptics prior to urinary catheterisation and effect on CAUTI” to explore points related to the GRADE Evidence to Framework approach – acceptability, feasibility cost, equity. In the conclusion, we have also explore the issue of antiseptics as a broad brush and made mention of evidence supporting chlorhexidine being	

		limited to one study. tempered the conclusion. We have also modified the conclusion to refer to issues with bacteriuria as an outcome.	
10	The outcome could be addressed more clearly.	Agree and with the changes in the previous two points above, we believe this is now addressed	See two points above.

VERSION 2 – REVIEW

REVIEWER	Solis-Trapala, I Keele, Institute for Applied Clinical Sciences
REVIEW RETURNED	21-Apr-2021

GENERAL COMMENTS	Thank you for addressing my comments from my review